

COMPREHENSIVE 54-GENE TARGETED NGS MYELOID MUTATION PANEL

Patient Name:	Deepak Khurana	Date of Exam:	September 14, 2026
MRN / Patient ID:	MG-GEN-2026-8822	Age / Gender:	48 Yrs / Male
Blood Group & Rh:	AB+ (Rh Positive)	Referring Doctor:	Dr. Hemant Malhotra, MD, FRCP (Medical Oncology)
Clinical Diagnosis:	De Novo Acute Myeloid Leukemia (Molecular Risk Stratification)	Specimen / Source:	Bone Marrow Genomic DNA (High Purity)

Gene	Genomic Alteration	Variant Allele Frequency	Clinical Tier	Pathogenicity
NPM1	c.860_863dupTCTG (p.Trp288fs)	38.4 % VAF	Tier I (Strong Actionable)	Pathogenic Mutation
FLT3-ITD	Internal Tandem Duplication (33bp)	Allelic Ratio: 0.12 (Low)	Tier I (Prognostic Marker)	Pathogenic (Low AR)
DNMT3A	c.2645G>A (p.Arg882His)	41.2 % VAF	Tier I (Preleukemic Clone)	Pathogenic Mutation
TP53	No Mutation Detected	< 1.0 % LOD	Tier I (Favorable)	Wild Type (Normal)
CEBPA	No Mutation Detected	< 1.0 % LOD	Tier I	Wild Type (Normal)
IDH1 / IDH2	No Mutation Detected	< 1.0 % LOD	Tier I	Wild Type (Normal)

CLINICAL INTERPRETATION & RECOMMENDATIONS:

- Next Generation Sequencing (Illumina NovaSeq 6000 at 1,500x mean coverage) identified NPM1 mutation.
- Co-existing FLT3-ITD mutation detected at LOW allelic ratio (AR 0.12 < 0.50).
- According to European LeukemiaNet (ELN 2022) criteria, NPM1-mutated with FLT3-ITD low represents FAVORABLE RISK.
- Recommendation: High probability of response to standard anthracycline induction; potential deferral of frontline BMT.

Dr. V. L. Ramprasad, Ph.D. (Chief Operating Officer)

Medical Technologist / Pathologist

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